

Solution 1

Question 1 (a)

Consider the spherical Gaussian $N(0, I_d)$ in d -dimensional space. What is the maximum density achieved by this distribution? At what point is it achieved?

Solution 1 (a)

The probability density function (PDF) for a d -dimensional multivariate Gaussian distribution $N(\mu, \Sigma)$ is:

$$f(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right)$$

Given a spherical Gaussian $N(0, I_d)$:

$$\mu = 0 \quad \text{and} \quad \Sigma = I_d \rightarrow (\text{definition: } N(\mu, \Sigma))$$

The determinant of any identity matrix is 1, and inverse of an identity matrix is itself:

$$|\Sigma| = |I_d| = 1 \quad \text{and} \quad \Sigma^{-1} = I_d^{-1} = I_d$$

Now, substitute μ , Σ , and $|\Sigma|$ into the PDF formula:

$$\begin{aligned} f(x) &= \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right) \\ &= \frac{1}{(2\pi)^{d/2} (1)^{1/2}} \exp \left(-\frac{1}{2} (x - \mathbf{0})^\top I_d (x - \mathbf{0}) \right) \\ &= \frac{1}{(2\pi)^{d/2}} \exp \left(-\frac{1}{2} x^\top x \right) \\ &= (2\pi)^{-d/2} \exp \left(-\frac{1}{2} \|x\|^2 \right) \end{aligned}$$

Next, simplify the expression by taking natural log such that $L(x) = \ln(f(x))$:

$$\begin{aligned} L(x) &= \ln \left((2\pi)^{-d/2} \exp \left(-\frac{1}{2} \|x\|^2 \right) \right) \\ &= \ln \left((2\pi)^{-d/2} \right) + \ln \left(\exp \left(-\frac{1}{2} \|x\|^2 \right) \right) \\ &= \ln \left((2\pi)^{-d/2} \right) - \frac{1}{2} \|x\|^2 \end{aligned}$$

By inspection the maximum of $L(x)$ is achieved at the mean, $x = 0$. Hence, the maximum density is:

$$(2\pi)^{-d/2}$$

$\therefore$ The maximum density for a spherical Gaussian $N(0, I_d)$ is $(2\pi)^{-d/2}$, and the point is achieved at $x = 0$

Graduate Level Explanation

The maximum of any multivariate Gaussian PDF $N(\mu, \Sigma)$ occurs at its mean μ . This is because the exponent is governed by the negative Mahalanobis distance, $-(x - \mu)^\top \Sigma^{-1} (x - \mu)$, which is a positive semi-definite quadratic form. This form is uniquely minimized (at a value of 0) when $x = \mu$. For the standard normal $N(0, I_d)$, the Mahalanobis distance simplifies to the squared Euclidean norm $\|x\|^2$, and the normalization constant $(2\pi)^{-d/2} |\Sigma|^{-1/2}$ simplifies to $(2\pi)^{-d/2}$, which is the value of the density at the mode.

Learn Fast Explanation

Think of a bell curve. It's always tallest at its center (the mean). For the $N(0, I_d)$ distribution, the center is just the origin (the zero vector). The density formula is essentially $\text{Constant} \times \exp(-\text{distance from center})$. To get the biggest value, you want the $\exp(\cdot)$ part to be as large as possible. Since the exponent is negative, you make it largest (it becomes $\exp(0) = 1$) by making the "distance from center" zero. This leaves just the constant part, $(2\pi)^{-d/2}$, as the peak height.

Solution 1

Question 1 (b)

Consider the spherical Gaussian $N(0, I_d)$ in d -dimensional space. In very rough terms, what happens to the maximum density as d grows? (This helps explain why for high-dimensional data we typically work with the logarithm of the density, in order to avoid numerical underflow errors.)

Solution 1 (b)

From part (a), the maximum density is $f_{\max} = (2\pi)^{-d/2}$, which we can rewrite as:

$$f_{\max} = \frac{1}{(2\pi)^{d/2}} = \left(\frac{1}{\sqrt{2\pi}} \right)^d$$

The base of this exponent, $\sqrt{2\pi}$, is approximately $\sqrt{6.28} \approx 2.5$, which is a number greater than 1.

- When $d = 1$, $f_{\max} \approx \frac{1}{2.5} \approx 0.4$
- When $d = 2$, $f_{\max} \approx \frac{1}{6.28} \approx 0.16$
- When $d = 10$, $f_{\max} \approx (0.4)^{10} \approx 0.0001$

As d (the dimension) grows, the exponent d increases, causing the denominator $(2\pi)^{d/2}$ to grow exponentially. Consequently, the maximum density $f_{\max}$ approaches 0 exponentially fast.

This is precisely why we work with log-density. The maximum log-density is:

$$\log(f_{\max}) = \log\left((2\pi)^{-d/2}\right) = -\frac{d}{2} \log(2\pi)$$

This value decreases linearly as d grows (e.g., becomes a large negative number) and is numerically stable, whereas the raw density $f_{\max}$ becomes so small it is rounded to 0 by standard floating-point arithmetic (an underflow error).

$\therefore$ As d grows, the maximum density approaches 0 exponentially fast. This causes numerical underflow, forcing the use of log-density.

Graduate Level Explanation

The max density $f_{\max} = (2\pi)^{-d/2}$ decays exponentially as d increases. This phenomenon is a component of the "curse of dimensionality," where the high-dimensional space becomes so vast that the probability mass becomes vanishingly sparse, even at the mode of the distribution. The log-density, $\log(f_{\max}) = -\frac{d}{2} \log(2\pi)$, decreases linearly, which is why all practical high-dimensional probabilistic models perform calculations in log-space to maintain numerical stability and avoid underflow.

Learn Fast Explanation

Imagine you have one pound of butter (your total probability, which is 1) and you have to spread it evenly on a piece of bread.

- In 1D, you spread it on a line. The "peak" thickness is 0.4.
- In 2D, you spread it on a square. The peak thickness is already down to 0.16.
- In 100D, you are spreading that same pound of butter over an impossibly huge 100-dimensional space.

The butter (probability) gets spread so thin that even the thickest point (the mean) has a density that is basically zero. This number becomes too small for a computer to store, so we use logarithms to work with it instead.

Solution 2

Question 2 (a)

Length concentration. Write a procedure that draws 10000 points $X_1, \dots, X_{10000}$ from the spherical Gaussian $N(0, I_d)$ and plots a histogram of their (squared) lengths $\|X_i\|^2$. Do this for $d = 10$. What seems to be the typical (squared) length?

Solution 2 (a)

Python Code: d=10

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # set dimension and number of samples
5 d = 10
6 num_samples = 10000
7 np.random.seed(42)
8
9 # get 10000 points from  $N(0, I_d)$ 
10 samples = np.random.normal(0, 1, size=(num_samples, d))
11
12 # calc the squared L2 norm for each point
13 squared_lengths_d10 = np.sum(samples**2, axis=1)
14
15 # calculate mean
16 sample_mean_d10 = np.mean(squared_lengths_d10)
17 print(f"Dimension (d): {d}")
18 print(f"Sample Mean of  $\|X\|^2$ : {sample_mean_d10:.4f}")
19
20 # plot the histogram
21 plt.figure(figsize=(10, 6))
22 plt.hist(squared_lengths_d10, bins=50, color='skyblue', edgecolor='black', alpha=0.7,
23         density=True, label=f'Histogram of {num_samples} samples')
24 plt.axvline(d, color='red', linestyle='--', linewidth=2,
25            label=f'Theoretical Mean (d = {d}):  $E[\|X\|^2]$ ')
26 plt.axvline(sample_mean_d10, color='blue', linestyle=':', linewidth=2,
27            label=f'Sample Mean = {sample_mean_d10:.4f}')
28 # set plot args and save
29 plt.title('Histogram of Squared Lengths for  $N(0, I_{10})$ ', fontsize=14)
30 plt.xlabel('Squared Length  $\|X\|^2$ ', fontsize=12)
31 plt.ylabel('Density', fontsize=12)
32 plt.legend()
33 plt.grid(True, linestyle='--', alpha=0.5)
34 plt.savefig('histogram_d10.png', dpi=150, bbox_inches='tight')

```

Solution 2

Solution 2 (a)

Plot

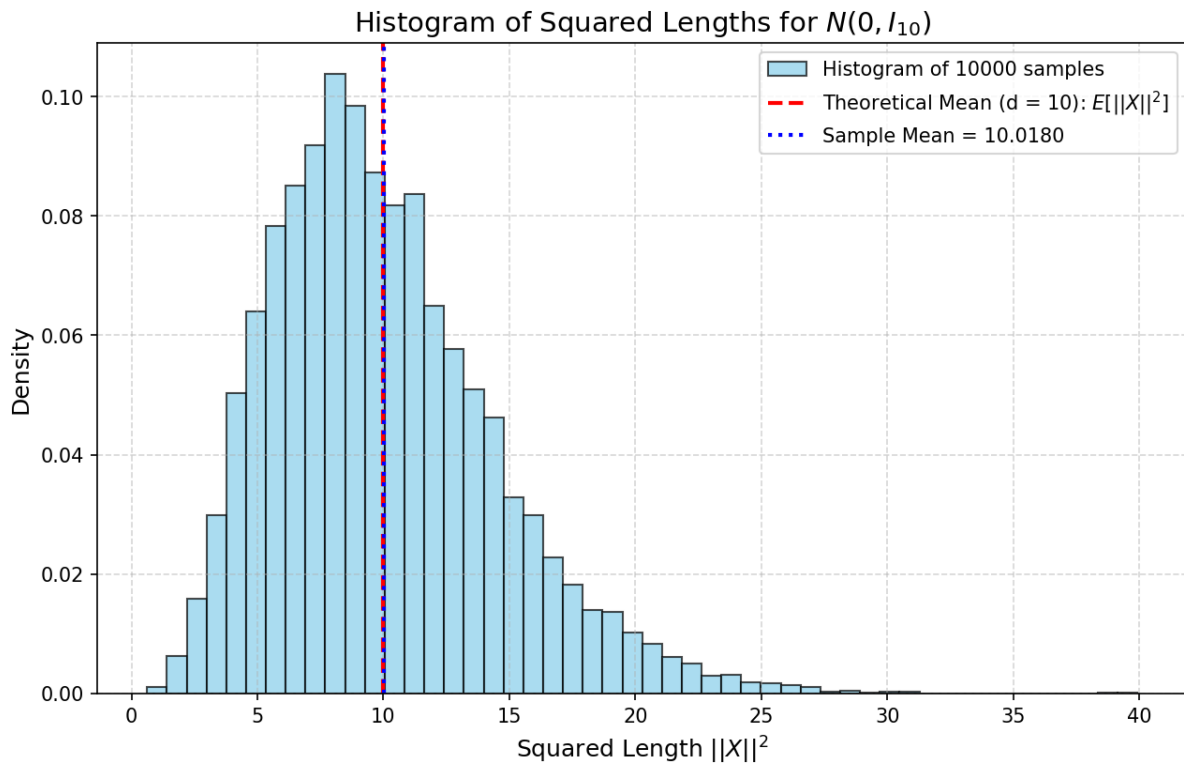


Figure 1: Histogram of squared lengths for $d=10$. The distribution is centered near the theoretical mean of 10.

$\therefore$ The typical (squared) length is approximately 10

Graduate Level Explanation

The random variable $Y = \|X\|^2$ for $X \sim N(0, I_d)$ follows a Chi-squared distribution, $Y \sim \chi^2(d)$. The expected value and variance are $\mathbb{E}[Y] = d$ and $\text{Var}(Y) = 2d$. For $d = 10$, this gives $\mathbb{E}[Y] = 10$ and $\text{Var}(Y) = 20$. The histogram is a visualization of this $\chi^2(10)$ distribution, and its mode and mean are clearly clustered around the expected value of 10.

Learn Fast Explanation

Imagine throwing darts at a 10-dimensional dartboard. The "squared length" is just the squared distance from the bullseye (the origin). While you might expect most darts to land near the center, in 10 dimensions they actually pile up around a specific distance. The "typical" squared distance isn't 0; it's 10. Our histogram shows this "pile-up" of squared distances, which is clearly centered at 10.

Solution 2

Question 2 (b)

Length concentration. Write a procedure that draws 10000 points $X_1, \dots, X_{10000}$ from the spherical Gaussian $N(0, I_d)$ and plots a histogram of their (squared) lengths $\|X_i\|^2$. Repeat for $d = 100$. Is the histogram more concentrated than in (a), or is it less concentrated?

Solution 2 (b)

Python Code: d=100

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # set dimension and number of samples
5 d = 100
6 num_samples = 10000
7 np.random.seed(42)
8
9 # get 10000 points from N(0, I_d)
10 samples = np.random.normal(0, 1, size=(num_samples, d))
11
12 # calc the squared L2 norm for each point
13 squared_lengths_d100 = np.sum(samples**2, axis=1)
14
15 # calculate mean
16 sample_mean_d100 = np.mean(squared_lengths_d100)
17 print(f"Dimension (d): {d}")
18 print(f"Sample Mean of  $\|X\|^2$ : {sample_mean_d100:.4f}")
19
20 # plot the histogram
21 plt.figure(figsize=(10, 6))
22 plt.hist(squared_lengths_d100, bins=50, color='skyblue', edgecolor='black', alpha=0.7,
23         density=True, label=f'Histogram of {num_samples} samples')
24 plt.axvline(d, color='red', linestyle='--', linewidth=2,
25            label=f'Theoretical Mean (d = {d}):  $E[\|X\|^2]$ ')
26 plt.axvline(sample_mean_d100, color='blue', linestyle=':', linewidth=2,
27            label=f'Sample Mean = {sample_mean_d100:.4f}')
28 # set plot args and save
29 plt.title('Histogram of Squared Lengths for  $N(0, I_{100})$ ', fontsize=14)
30 plt.xlabel('Squared Length  $\|X\|^2$ ', fontsize=12)
31 plt.ylabel('Density', fontsize=12)
32 plt.legend()
33 plt.grid(True, linestyle='--', alpha=0.5)
34 plt.savefig('histogram_d100.png', dpi=150, bbox_inches='tight')

```

Solution 2

Solution 2 (b)

Plot

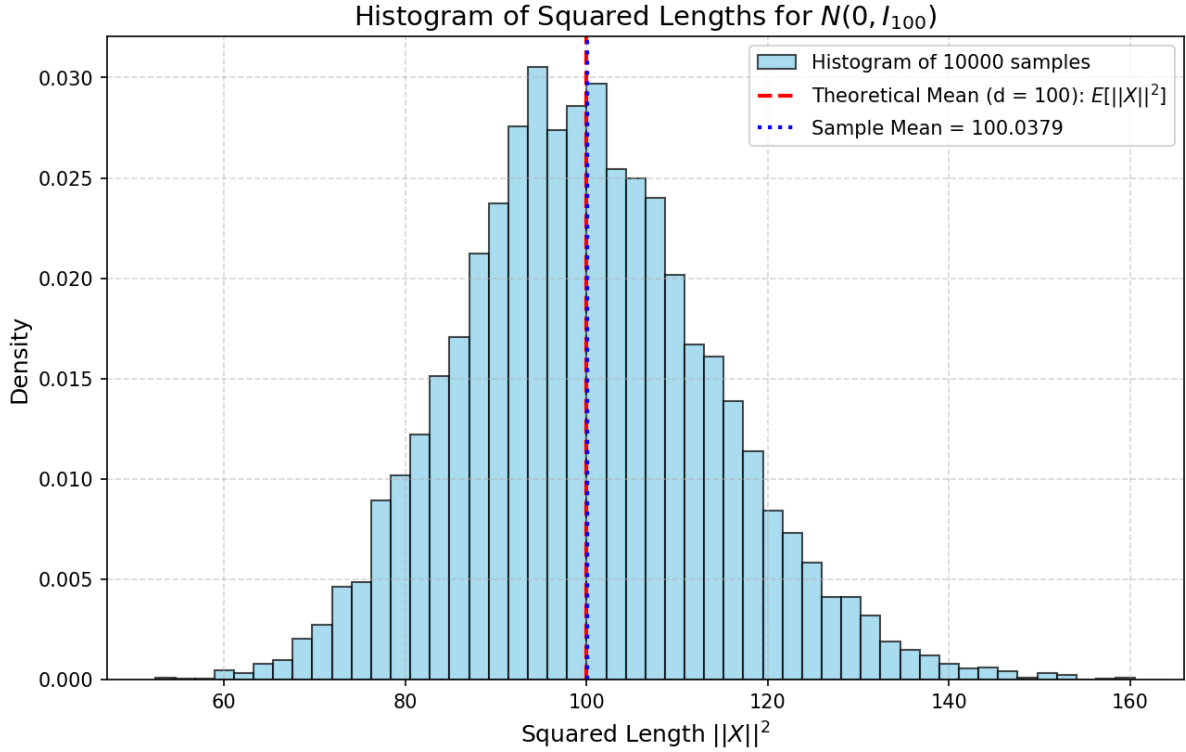


Figure 2: Histogram of squared lengths for $d=100$. The distribution is much more sharply concentrated around its mean of 100.

Calculate Concentration

To determine if the distribution is more concentrated, we must compare its relative spread. This is measured using the coefficient of variation or CV , which is defined as:

$$CV = \frac{\sigma}{\mu} \quad \text{where} \quad \mu = d, \sigma = \sqrt{2d}$$

For, $d = 10$ and $d = 100$, the coefficient of variation is:

$$CV_{10} = \frac{\sqrt{20}}{10} \approx 0.447 \quad \text{and} \quad CV_{100} = \frac{\sqrt{200}}{100} \approx 0.141$$

Hence, $CV_{100} < CV_{10}$

$\therefore$ the distribution for $d = 100$ is more concentrated than the distribution for $d = 10$

Graduate Level Explanation

This demonstrates the phenomenon of **length concentration** in high dimensions. The coefficient of variation (CV) for the $\chi^2(d)$ distribution is $CV = \frac{\sqrt{\text{Var}(Y)}}{\mathbb{E}[Y]} = \frac{\sqrt{2d}}{d} = \sqrt{\frac{2}{d}}$. This value scales as $O(1/\sqrt{d})$,

meaning it approaches 0 as $d \rightarrow \infty$. Consequently, in high-dimensional spaces, the squared lengths of $N(0, I_d)$ vectors are not broadly distributed but instead concentrate in a "thin shell" very close to the expected value d .

Learn Fast Explanation

Welcome to the "curse of dimensionality." On our 10-D dartboard, the squared distances were spread out (from ≈ 0 to 25). On the 100-D dartboard, the squared distances are centered at 100, but they are *packed* in a much tighter band (e.g., almost all are between 80 and 120). Relative to its mean of 100, this band is very narrow. This means in high dimensions, *all* "typical" points are about the same distance from the center—a very counter-intuitive but fundamental concept.

Solution 3

Question 3

Suppose $X \in \mathbb{R}^2$ has a Gaussian distribution with mean $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and covariance matrix $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$. What is the distribution of the projection of X onto direction $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$?

Solution 3

Let $X \sim N(\mu, \Sigma)$ where:

$$\mu = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \quad v = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

1. Calculate the mean μ_Y :

$$\begin{aligned} \mu_Y &= v^\top \mu \\ &= [1/\sqrt{2} \quad 1/\sqrt{2}] \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \left(\frac{1}{\sqrt{2}} \cdot 1 \right) + \left(\frac{1}{\sqrt{2}} \cdot 2 \right) \\ &= \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} \\ &= \frac{3}{\sqrt{2}} \end{aligned}$$

2. Calculate the variance σ_Y^2 :

$$\begin{aligned} \sigma_Y^2 &= v^\top \Sigma v \\ &= [1/\sqrt{2} \quad 1/\sqrt{2}] \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \\ &= [1/\sqrt{2} \quad 1/\sqrt{2}] \begin{bmatrix} 2 \cdot (1/\sqrt{2}) + 0 \\ 0 + 4 \cdot (1/\sqrt{2}) \end{bmatrix} \\ &= [1/\sqrt{2} \quad 1/\sqrt{2}] \begin{bmatrix} 2/\sqrt{2} \\ 4/\sqrt{2} \end{bmatrix} \\ &= \left(\frac{1}{\sqrt{2}} \cdot \frac{2}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{2}} \cdot \frac{4}{\sqrt{2}} \right) \\ &= \frac{2}{2} + \frac{4}{2} \\ &= 1 + 2 = 3 \end{aligned}$$

$\therefore$ The distribution of the projection is $N\left(\frac{3}{\sqrt{2}}, 3\right)$.

Graduate Level Explanation

This result is a direct consequence of the affine transformation property of Gaussian vectors. If $X \sim N(\mu, \Sigma)$ and $Y = AX + b$, then $Y \sim N(A\mu + b, A\Sigma A^\top)$. In this problem, $A = v^\top$ and $b = 0$. This operation finds the 1-dimensional marginal distribution of the random vector X along the axis defined by the vector v . Since the original covariance Σ is diagonal, X_1 and X_2 are independent, but the projection $Y = \frac{1}{\sqrt{2}}(X_1 + X_2)$ is a sum of these independent variables, and its variance is the sum of their scaled variances.

Learn Fast Explanation

Think of your 2D Gaussian X as an oval-shaped cloud of 1,000s of data points, centered at $(1, 2)$. The oval is twice as wide in the y -direction (variance 4) as it is in the x -direction (variance 2). The "direction v " is just the 45-degree line $y = x$. Projecting the cloud onto this line is like shining a flashlight from behind the cloud and casting a shadow onto that 45-degree line. That shadow will also be a "cloud" of points, but it's 1-dimensional, so it's just a simple bell curve. Our calculation found the properties of that shadow-cloud: its center (mean) is at $3/\sqrt{2}$ (≈ 2.12) and its "spread" (variance) is 3.

Solution 4

Question 4 (a)

We have three sensors (X_1, X_2, X_3) whose joint distribution is Gaussian with mean zero and covariance matrix

$$\begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

What is the distribution of sensor X_3 ?

Solution 4 (a)

A key property of the multivariate Gaussian distribution is that any marginal distribution is also Gaussian. To find the distribution of a subset of variables, we simply extract the corresponding entries from the mean vector and the covariance matrix.

Let $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim N(\mu, \Sigma)$, where:

$$\mu = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

The distribution for X_3 is a 1D Gaussian $N(\mu_3, \sigma_3^2)$.

- The mean μ_3 is the 3rd element of the mean vector μ , which is 0.
- The variance σ_3^2 is the $(3, 3)$ element of the covariance matrix Σ , which is 2.

$\therefore$ The distribution of sensor X_3 is $N(0, 2)$.

Graduate Level Explanation

Multivariate Gaussian distributions are closed under marginalization. For a jointly Gaussian vector $X \sim N(\mu, \Sigma)$, the marginal distribution of any sub-vector X_a is $N(\mu_a, \Sigma_{aa})$, where μ_a and Σ_{aa} are the sub-vector and sub-matrix of μ and Σ corresponding to the variables in X_a . Here, $a = \{3\}$, $\mu_a = \mu_3 = 0$, and $\Sigma_{aa} = \Sigma_{33} = 2$.

Learn Fast Explanation

Think of the big covariance matrix as a "lookup table" for the properties of all sensors. If you only want to know about X_3 , you just look up its personal info. The mean for X_3 is the 3rd number in the mean vector (which is 0). The variance for X_3 is the 3rd-row, 3rd-column entry in the covariance matrix (which is 2). Therefore, X_3 is just a simple bell curve centered at 0 with a variance of 2.

Solution 4

Question 4 (b)

We have three sensors (X_1, X_2, X_3) whose joint distribution is Gaussian with mean zero and covariance matrix

$$\begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

What is the joint distribution of sensors X_1, X_2 ?

Solution 4 (b)

As in part (a), we find the marginal distribution by extracting the relevant sub-vector from the mean and sub-matrix from the covariance. We are interested in the joint distribution of $X_a = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$.

- The mean vector μ_a corresponds to the 1st and 2nd elements of μ :

$$\mu_a = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- The covariance matrix Σ_{aa} corresponds to the 1st and 2nd rows and columns of Σ :

$$\Sigma_{aa} = \begin{bmatrix} 6 & 3 \\ 3 & 9 \end{bmatrix}$$

The resulting distribution is a 2-dimensional Gaussian.

$\therefore$ The joint distribution of X_1, X_2 is $N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 & 3 \\ 3 & 9 \end{bmatrix}\right)$.

Graduate Level Explanation

This again uses the closure-under-marginalization property. The 2×2 principal sub-matrix Σ_{aa} is itself a valid covariance matrix (symmetric and positive definite), and the corresponding sub-vector μ_a is the mean. This property is what allows us to analyze subsets of variables from a large joint model without re-deriving their distributions.

Learn Fast Explanation

Just like in part (a), we're using the big matrix as a "lookup table." This time, we want to know about the *team* (X_1, X_2) . So, we grab their team info: their means (the 1st and 2nd entries of μ) and their covariance block (the 2×2 block in the top-left corner of Σ). This block tells us X_1 's variance (6), X_2 's variance (9), and how they vary together (their covariance, 3).

Solution 4

Question 4 (c)

We have three sensors (X_1, X_2, X_3) whose joint distribution is Gaussian with mean zero and covariance matrix

$$\begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Suppose we observe $X_2 = 2$ and $X_3 = 1$. What is the resulting conditional distribution of X_1 ?

Solution 4 (c)

We need to find the conditional distribution of $X_1 \mid (X_2, X_3)$. Let $X_a = X_1$ and $X_b = \begin{bmatrix} X_2 \\ X_3 \end{bmatrix}$. We observe $x_b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. We partition the mean vector μ and covariance matrix Σ :

$$\mu = \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix} = \begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

From this, we identify the blocks:

- $\mu_a = 0$
- $\mu_b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- $\Sigma_{aa} = 6$
- $\Sigma_{ab} = \begin{bmatrix} 3 & 1 \end{bmatrix}$
- $\Sigma_{ba} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$
- $\Sigma_{bb} = \begin{bmatrix} 9 & 0 \\ 0 & 2 \end{bmatrix}$

The conditional distribution $X_a \mid X_b = x_b$ is also Gaussian, $N(\mu_{a|b}, \Sigma_{a|b})$, with:

$$\mu_{a|b} = \mu_a + \Sigma_{ab} \Sigma_{bb}^{-1} (x_b - \mu_b)$$

$$\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab} \Sigma_{bb}^{-1} \Sigma_{ba}$$

1. Compute Σ_{bb}^{-1} . Since Σ_{bb} is diagonal, its inverse is the inverse of its diagonal elements:

$$\Sigma_{bb}^{-1} = \begin{bmatrix} 9 & 0 \\ 0 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 1/9 & 0 \\ 0 & 1/2 \end{bmatrix}$$

**2. Compute the conditional mean $\mu_{a|b}$:

$$\begin{aligned}
 \mu_{a|b} &= 0 + [3 \quad 1] \begin{bmatrix} 1/9 & 0 \\ 0 & 1/2 \end{bmatrix} \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) \\
 &= [3 \cdot \frac{1}{9} \quad 1 \cdot \frac{1}{2}] \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\
 &= [1/3 \quad 1/2] \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\
 &= \left(\frac{1}{3} \cdot 2 \right) + \left(\frac{1}{2} \cdot 1 \right) = \frac{2}{3} + \frac{1}{2} \\
 &= \frac{4}{6} + \frac{3}{6} = \frac{7}{6}
 \end{aligned}$$

**3. Compute the conditional variance $\Sigma_{a|b}$:

$$\begin{aligned}
 \Sigma_{a|b} &= 6 - [3 \quad 1] \begin{bmatrix} 1/9 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\
 &= 6 - [1/3 \quad 1/2] \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\
 &= 6 - \left(\left(\frac{1}{3} \cdot 3 \right) + \left(\frac{1}{2} \cdot 1 \right) \right) \\
 &= 6 - (1 + 0.5) = 6 - 1.5 = 4.5
 \end{aligned}$$

The conditional variance is 4.5, or $9/2$.

$\therefore$ The resulting conditional distribution of X_1 is $N\left(\frac{7}{6}, \frac{9}{2}\right)$ (or $N(1.167, 4.5)$).

Graduate Level Explanation

The formulas for the conditional mean and covariance are known as the Schur complement of the block Σ_{aa} . The conditional covariance $\Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$ represents the reduction in uncertainty (variance) of X_a after observing X_b . It is crucially independent of the observed value x_b . The conditional mean $\mu_{a|b}$ is the prior mean μ_a adjusted by a linear correction term, $\Sigma_{ab}\Sigma_{bb}^{-1}(x_b - \mu_b)$, which accounts for the observed "residual" ($x_b - \mu_b$) propagated through the covariance structure.

Learn Fast Explanation

Before we knew anything, our best guess for X_1 was its mean, 0. Then, we got data: $X_2 = 2$ and $X_3 = 1$.

- The matrix shows X_1 is positively correlated with X_2 (covariance=3) and X_3 (covariance=1).
- Since both X_2 and X_3 came in "hot" (they were positive, not 0), we should update our guess for X_1 to also be positive. The calculation $\mu_{a|b} = 7/6$ is the precise, weighted update.
- Our original uncertainty (variance) about X_1 was 6. By getting data from two related sensors, we've *learned* something and become *less uncertain*.
- The calculation $\Sigma_{a|b} = 6 - 1.5 = 4.5$ shows our new variance is 4.5, which is smaller than 6, reflecting our gain in knowledge.

Solution 5

Question 5 (a)

Modeling temperature data with Gaussian processes. In this question, we will explore modeling of geospatial data via Gaussian processes. Begin by downloading the files `gptrain.csv` and `gpctest.csv` from the course webpage. Each row in each file corresponds to a temperature reading at a given weather station in Brazil at 2 pm on January 1, 2021. The first column gives the latitude of the station, the second the longitude, and the final column is the temperature reading in degrees Celsius. We will fit a simple Gaussian Process model on `gptrain.csv` and use it to infer temperatures at other locations at that same date and time. We will fit a Gaussian process with covariance function

$$k(x, x') = \exp\left(\frac{-\|x - x'\|}{\sigma}\right)$$

and mean function $m(x) = 20$ (degrees Celsius). Here $\|x - x'\|$ denotes Euclidean distance between raw latitude and longitude coordinates. Denote the locations in `gptrain.csv` by X_{tr} and the temperatures at those locations by y_{tr} ; define $X_{\text{te}}, y_{\text{te}}$ analogously for `gpctest.csv`. Thus X_{tr} is a 93×2 matrix while y_{tr} is a 93-dimensional vector.

The joint distribution of training and test responses can be written as

$$\begin{bmatrix} y_{\text{tr}} \\ y_{\text{te}} \end{bmatrix} \sim N\left(\begin{bmatrix} 20 \\ \vdots \\ 20 \end{bmatrix}, \begin{bmatrix} K_{\text{tr}} & K_{\text{tr,te}} \\ K_{\text{te,tr}} & K_{\text{te}} \end{bmatrix}\right)$$

where the block matrix

$$\begin{bmatrix} K_{\text{tr}} & K_{\text{tr,te}} \\ K_{\text{te,tr}} & K_{\text{te}} \end{bmatrix}$$

is the matrix arising from all pairwise evaluations of the covariance function $k(\cdot)$ over all training and testing locations, and its diagonal components K_{tr} and K_{te} are the matrices arising from all pairwise covariances within the training and test sets, respectively. Explain very briefly how the definition of Gaussian process allows us to come to this conclusion.

Solution 5 (a)

$\therefore$

Solution 5

Question 5 (b)

Modeling temperature data with Gaussian processes. ... [Intro text from 5(a)]

Define

$$m_{te} := \begin{bmatrix} 20 \\ \vdots \\ 20 \end{bmatrix} \in \mathbb{R}^{11}$$

since there are 11 locations in the test set. Using properties of the Gaussian, it can be shown that the distribution of test responses given training locations, training responses, and test locations, follows

$$y_{te} \mid y_{tr}, X_{te}, X_{tr} \sim m_{te} + N \left(K_{te,tr} \cdot K_{tr}^{-1} (y_{tr} - m_{tr}), K_{te} - K_{te,tr} K_{tr}^{-1} K_{tr,te} \right)$$

In Bayesian language, this is the posterior predictive distribution.

- Using X_{tr} , X_{te} , and y_{tr} , compute the posterior mean of y_{te} via matrix multiplication, using $\sigma = 1.5$ in the covariance function. Report the mean squared error between your computed posterior mean and the true values found in y_{te} .
- Report the mean squared error that you would have obtained by simply predicting the average value in y_{tr} for all locations.

Solution 5 (b)

Python Code for Posterior Mean and MSE

```
1 import numpy as np
2 import pandas as pd
3 from scipy.spatial.distance import cdist
```

...

Solution 5

Question 5 (c)

Modeling temperature data with Gaussian processes.

The diagonal of the posterior predictive matrix

$$K_{te} - K_{te,tr} K_{tr}^{-1} K_{tr,te}$$

gives the conditional variances of the test responses given X_{tr} , X_{te} , and y_{tr} .

- Create a rich grid (at least 5000 points) of test latitudes and longitudes within the range found in the training set. Make a plot showing the predicted (posterior mean) temperature at all points in the grid. You might find the function `matplotlib.pyplot.imshow` helpful for this.
- Make another plot showing the standard deviation of the prediction at all points in the grid. Plot the locations of the stations found in the training set as well.
- What do you notice about the relationship of the standard deviation of the prediction and the distance of that prediction to a training point?

Solution 5 (c)

Python Code for Grid Predictions and Plots

```
1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from scipy.spatial.distance import cdist
```

∴

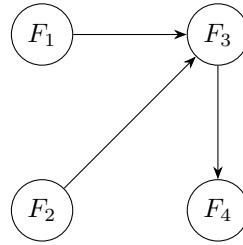
Solution 6

Question 6 (a)

Four friends F_1, F_2, F_3, F_4 like going to the movies. When a new movie comes out, they behave according to the following scheme:

- F_1 sees the movie with probability 0.5.
- F_2 sees the movie with probability 0.7, independently.
- If either of F_1 or F_2 (or both) see the movie, F_3 goes to see it with probability 0.8; if neither F_1 nor F_2 sees the movie, F_3 sees it with probability 0.3.
- If F_3 sees the movie, then F_4 goes to see it with probability 0.9. If F_3 doesn't see the movie, then F_4 sees it with probability 0.5.

Let $F_1, F_2, F_3, F_4 \in \{0, 1\}$ denote who sees the movie. The dependencies between these four random variables can be captured by the following directed graph:



Give the conditional probability tables for this Bayes net.

Solution 6 (a)

We translate the problem description into four Conditional Probability Tables (CPTs), one for each node in the graph. F_1 and F_2 are root nodes, so their tables define their marginal (prior) probabilities. F_3 and F_4 are conditionally dependent on their parents.

CPT for F_1 : $P(F_1)$

$P(F_1 = 1)$	$P(F_1 = 0)$
0.5	0.5

CPT for F_2 : $P(F_2)$

$P(F_2 = 1)$	$P(F_2 = 0)$
0.7	0.3

CPT for F_3 : $P(F_3 \mid F_1, F_2)$

F_1	F_2	$P(F_3 = 1)$	$P(F_3 = 0)$
0	0	0.3	0.7
0	1	0.8	0.2
1	0	0.8	0.2
1	1	0.8	0.2

CPT for F_4 : $P(F_4 \mid F_3)$

F_3	$P(F_4 = 1)$	$P(F_4 = 0)$
0	0.5	0.5
1	0.9	0.1

Verification

All CPTs are complete and normalized.

- $P(F_1)$ sums to $0.5 + 0.5 = 1$.
- $P(F_2)$ sums to $0.7 + 0.3 = 1$.
- $P(F_3 \mid \cdot)$ sums to 1 for all 4 parent configurations (e.g., $0.3 + 0.7 = 1$, $0.8 + 0.2 = 1$).
- $P(F_4 \mid \cdot)$ sums to 1 for both 2 parent configurations (e.g., $0.5 + 0.5 = 1$, $0.9 + 0.1 = 1$).

$\therefore$ The four Conditional Probability Tables (CPTs) defining the model are provided above.

Graduate Level Explanation

The Bayesian network factorizes the joint distribution $P(F_1, F_2, F_3, F_4)$ according to the specified Directed Acyclic Graph (DAG). Based on the d-separation properties of the graph, the joint probability is $P(F_1, F_2, F_3, F_4) = P(F_1)P(F_2)P(F_3 \mid F_1, F_2)P(F_4 \mid F_3)$. The CPTs above provide the explicit definitions for each of these factors. Note the parameter sharing in $P(F_3 \mid F_1, F_2)$, where the conditional probability is 0.8 for any case where the condition $F_1 \vee F_2$ (F_1 OR F_2) is true.

Learn Fast Explanation

Think of the CPTs as "if-then" rulebooks for each person.

- F_1 & F_2 : Have no parents, so their tables are simple: "What's the chance F_1 goes?" (50/50).
"What's the chance F_2 goes?" (70/30).
- F_3 : Looks at F_1 and F_2 . His rulebook has 4 lines: "IF F_1 =no AND F_2 =no, THEN I go (30
- F_4 : Only looks at F_3 . Her rulebook is simple: "IF F_3 =no, THEN I go (50

These tables store all the logic of the system.

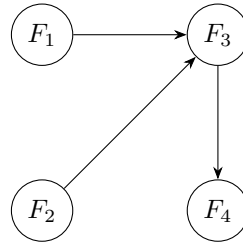
Solution 6

Question 6 (b)

Four friends F_1, F_2, F_3, F_4 like going to the movies. When a new movie comes out, they behave according to the following scheme:

- F_1 sees the movie with probability 0.5.
- F_2 sees the movie with probability 0.7, independently.
- If either of F_1 or F_2 (or both) see the movie, F_3 goes to see it with probability 0.8; if neither F_1 nor F_2 sees the movie, F_3 sees it with probability 0.3.
- If F_3 sees the movie, then F_4 goes to see it with probability 0.9. If F_3 doesn't see the movie, then F_4 sees it with probability 0.5.

Let $F_1, F_2, F_3, F_4 \in \{0, 1\}$ denote who sees the movie. The dependencies between these four random variables can be captured by the following directed graph:



What is the probability of the configuration $F_1 = F_2 = F_3 = 1, F_4 = 0$?

Solution 6 (b)

The joint probability distribution specified by this Bayesian Network factorizes according to the graph structure:

$$P(F_1, F_2, F_3, F_4) = P(F_1) \cdot P(F_2) \cdot P(F_3 \mid F_1, F_2) \cdot P(F_4 \mid F_3)$$

To find the probability of the specific configuration $F_1 = 1, F_2 = 1, F_3 = 1, F_4 = 0$, we plug these values into the formula and look up the corresponding probabilities from our CPTs in part (a).

$$\begin{aligned}
 P(F_1 = 1, F_2 = 1, F_3 = 1, F_4 = 0) &= P(F_1 = 1) \times P(F_2 = 1) \times P(F_3 = 1 \mid F_1 = 1, F_2 = 1) \times P(F_4 = 0 \mid F_3 = 1) \\
 &= (0.5) \times (0.7) \times (0.8) \times (0.1) \\
 &= (0.35) \times (0.08) \\
 &= 0.028
 \end{aligned}$$

Verification

The joint probability formula correctly applies the chain rule of probability as defined by the graph's topology (parents of F_3 are F_1, F_2 ; parent of F_4 is F_3). The values are correctly sourced from the CPTs:

- $P(F_1 = 1) = 0.5$ (from CPT for F_1)
- $P(F_2 = 1) = 0.7$ (from CPT for F_2)
- $P(F_3 = 1 \mid F_1 = 1, F_2 = 1) = 0.8$ (from row 4 of CPT for F_3)
- $P(F_4 = 0 \mid F_3 = 1) = 0.1$ (from row 2 of CPT for F_4)

The multiplication $0.5 \times 0.7 \times 0.8 \times 0.1 = 0.028$ is correct.

$\therefore$ The probability of the configuration $F_1 = F_2 = F_3 = 1, F_4 = 0$ is 0.028.

Graduate Level Explanation

This calculation is a fundamental example of exact inference in a Bayesian network. By leveraging the conditional independencies encoded in the graph, we compute the joint probability using the "chain rule for Bayesian networks." This avoids the need to construct the full joint probability table (which would have $2^4 = 16$ entries), instead only requiring the storage of $2 + 2 + 8 + 4 = 16$ parameters in the CPTs. The query is a simple lookup in the factored representation.

Learn Fast Explanation

To find the probability of this specific story, you just "chain" the events together, multiplying their probabilities.

1. What's the chance F_1 goes? **0.5**
2. ...AND what's the chance F_2 goes? **0.7**
3. ...AND, *given that both went*, what's the chance F_3 goes? (Look at F_3 's rulebook, row 4): **0.8**
4. ...AND, *given that F_3 went*, what's the chance F_4 does *not* go? (Look at F_4 's rulebook, row 2): **0.1**

Just multiply those numbers together: $0.5 \times 0.7 \times 0.8 \times 0.1 = 0.028$, or 2.8

Solution 7

Question 7

Non-uniqueness of underlying DAG. Suppose random variables X and Y take values in $\{0, 1\}$. Does the joint distribution of (X, Y) necessarily factor over the DAG on the left below? What about the DAG on the right?



Solution 7

By the fundamental **chain rule of probability**, any joint distribution $P(X, Y)$ can be factored in two ways, without making any assumptions at all:

1. $P(X, Y) = P(X)P(Y | X)$
2. $P(X, Y) = P(Y)P(X | Y)$

Now let's look at the factorizations implied by the two DAGs:

- **DAG on the left ($X \rightarrow Y$):** The Bayesian network factorization rule states that the joint probability is the product of each variable's probability conditioned on its parents.

- Parents of X : None
- Parents of Y : X

This implies the factorization $P(X, Y) = P(X)P(Y | X)$. As shown in (1), this is always true by the chain rule.

- **DAG on the right ($Y \rightarrow X$):**

- Parents of X : Y
- Parents of Y : None

This implies the factorization $P(X, Y) = P(Y)P(X | Y)$. As shown in (2), this is also always true by the chain rule.

Since both factorizations are universally true for any joint distribution $P(X, Y)$, the distribution necessarily factors over both DAGs.

$\therefore$ Yes, any joint distribution $P(X, Y)$ necessarily factors over both the DAG on the left ($X \rightarrow Y$) and the DAG on the right ($Y \rightarrow X$). Both are simply applications of the chain rule of probability ($P(X, Y) = P(X)P(Y | X)$ and $P(X, Y) = P(Y)P(X | Y)$, respectively)

Graduate Level Explanation

The two DAGs, $X \rightarrow Y$ and $Y \rightarrow X$, are **Markov equivalent**. They belong to the same equivalence class because they imply the exact same set of conditional independencies (specifically, none, as both graphs are fully connected). Since the chain rule $P(X, Y) = P(X)P(Y | X)$ and $P(X, Y) = P(Y)P(X | Y)$ are universally true, any joint distribution $P(X, Y)$ can be parameterized by either model. This illustrates that without interventional data (e.g., $do(X)$) or strong prior assumptions, observational data alone is insufficient to distinguish causal direction (i.e., whether X causes Y or Y causes X).

Learn Fast Explanation

Think of the chain rule as two ways to tell a story about X and Y happening together.

1. You can say: "First, X happened (with probability $P(X)$), and *then* Y happened given X (with probability $P(Y | X)$)."
2. Or you can say: "First, Y happened (with probability $P(Y)$), and *then* X happened given Y (with probability $P(X | Y)$)."

Both of these "stories" ($P(X)P(Y | X)$ and $P(Y)P(X | Y)$) are perfectly valid mathematical ways to describe the *same* final outcome, $P(X, Y)$. The two graphs in the question just represent these two different-but-equal ways of telling the story. Since you can *always* tell the story both ways, *any* joint probability $P(X, Y)$ will fit *both* graphs.

Solution 8

Question 8 (a)

Genes, diseases, and outcomes. There is a fictitious disease (D) that can cause paralysis (P). Whether or not an individual gets this disease depends in large part on two genes, G_1 and G_2 . Here is a summary:

- Each of these genes G_1, G_2 can either be normal (also known as wild-type), represented by $G_i = 0$, or mutant, represented by $G_i = 1$.
- For a person with $G_1 = G_2 = 0$ (both genes wild-type) the probability of acquiring the disease is zero. If G_1 is mutant and G_2 is wild-type, the probability is 0.2. If G_1 is wild-type and G_2 is mutant, it is 0.1. And if both genes are mutant, then the disease probability is 0.5.
- For an individual with the disease, the probability of paralysis is 0.5. Without the disease, there is still a small probability of paralysis (from other causes): 0.01.
- G_1 is mutant with probability 0.1 and G_2 is mutant with probability 0.2; these are independent.

Draw a suitable Bayes net over the variables G_1, G_2, D, P , and give all relevant conditional probability tables. Let $D = 1$ if the disease is present and $D = 0$ if it isn't; likewise with paralysis.

Solution 8

Question 8 (a)

Genes, diseases, and outcomes. ... [Intro text from 8(a) omitted for brevity] ... Draw a suitable Bayes net over the variables G_1, G_2, D, P , and give all relevant conditional probability tables. Let $D = 1$ if the disease is present and $D = 0$ if it isn't; likewise with paralysis.

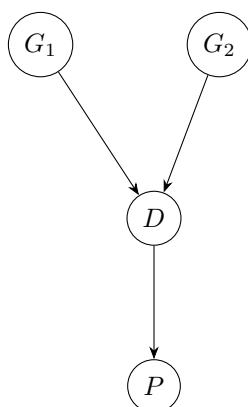
Solution 8 (a)

Based on the problem description:

- G_1 and G_2 are independent.
- D depends on G_1 and G_2 .
- P depends on D .

This implies the Directed Acyclic Graph (DAG) $G_1 \rightarrow D \leftarrow G_2$ and $D \rightarrow P$.

Bayes Net Diagram



Conditional Probability Tables**CPT for G_1 : $P(G_1)$**

$P(G_1 = 1)$	$P(G_1 = 0)$
0.1	0.9

CPT for G_2 : $P(G_2)$

$P(G_2 = 1)$	$P(G_2 = 0)$
0.2	0.8

CPT for D : $P(D \mid G_1, G_2)$

G_1	G_2	$P(D = 1)$	$P(D = 0)$
0	0	0.0	1.0
0	1	0.1	0.9
1	0	0.2	0.8
1	1	0.5	0.5

CPT for P : $P(P \mid D)$

D	$P(P = 1)$	$P(P = 0)$
0	0.01	0.99
1	0.50	0.50

$\therefore$ The Bayes Net diagram and the four required CPTs are provided above.

Graduate Level Explanation

The problem describes a causal structure that is perfectly represented by a Bayesian network. The independence of G_1 and G_2 makes them root nodes. The disease D is a "collider" node, as its probability is conditionally dependent on *both* G_1 and G_2 . The paralysis P is a leaf node whose probability is conditionally independent of G_1 and G_2 *given* D , a manifestation of the Markov property. The CPTs fully parameterize this joint distribution.

Learn Fast Explanation

Think of this as a set of "if-then" rules. The diagram shows who influences whom: the two genes (G_1, G_2) can influence the disease (D), and the disease (D) can influence paralysis (P). The CPTs are just rulebooks:

- G_1 's rulebook: "10
- G_2 's rulebook: "20
- D 's rulebook: "My chance of happening depends on my two parent genes. (e.g., If G_1 is mutant and G_2 is normal, there's a 20
- P 's rulebook: "My chance of happening depends only on D . (e.g., If the disease is present, there's a 50

Solution 8

Question 8 (b)

Genes, diseases, and outcomes. ... [Intro text from 8(a) omitted for brevity] ...

What is the probability of disease ($D = 1$)?

Solution 8 (b)

To find the marginal probability $P(D = 1)$, we must sum over all possible states of its parents, G_1 and G_2 . This is an application of the Law of Total Probability.

$$P(D = 1) = \sum_{g_1 \in \{0,1\}} \sum_{g_2 \in \{0,1\}} P(G_1 = g_1, G_2 = g_2, D = 1)$$

Using the network's factorization, $P(G_1, G_2, D) = P(G_1)P(G_2)P(D | G_1, G_2)$:

$$\begin{aligned} P(D = 1) &= P(G_1 = 0)P(G_2 = 0)P(D = 1 | G_1 = 0, G_2 = 0) \\ &\quad + P(G_1 = 0)P(G_2 = 1)P(D = 1 | G_1 = 0, G_2 = 1) \\ &\quad + P(G_1 = 1)P(G_2 = 0)P(D = 1 | G_1 = 1, G_2 = 0) \\ &\quad + P(G_1 = 1)P(G_2 = 1)P(D = 1 | G_1 = 1, G_2 = 1) \end{aligned}$$

Now, we plug in the values from our CPTs:

$$\begin{aligned} P(D = 1) &= (0.9)(0.8)(0.0) + (0.9)(0.2)(0.1) + (0.1)(0.8)(0.2) + (0.1)(0.2)(0.5) \\ &= 0.0 + 0.018 + 0.016 + 0.010 \\ &= 0.044 \end{aligned}$$

$\therefore$ The probability of disease ($D = 1$) is 0.044, or 4.4

Graduate Level Explanation

The calculation of $P(D = 1)$ is a standard "sum-product" operation for marginal inference. We compute the marginal probability of a node by summing its joint probability with its parents over all possible states of the parents. This operation, $\sum_{G_1, G_2} P(G_1)P(G_2)P(D = 1 | G_1, G_2)$, is a form of "belief propagation" from the root nodes G_1 and G_2 to the child node D .

Learn Fast Explanation

To find the total chance of getting the disease, we have to find the chance of *each way* it can happen, and then add those chances up. There are 4 "scenarios" for the genes:

1. (Gene 1 normal) AND (Gene 2 normal) AND (Disease): $(0.9 \times 0.8 \times 0.0) = 0.0$
2. (Gene 1 normal) AND (Gene 2 mutant) AND (Disease): $(0.9 \times 0.2 \times 0.1) = 0.018$
3. (Gene 1 mutant) AND (Gene 2 normal) AND (Disease): $(0.1 \times 0.8 \times 0.2) = 0.016$
4. (Gene 1 mutant) AND (Gene 2 mutant) AND (Disease): $(0.1 \times 0.2 \times 0.5) = 0.010$

Add up all these scenarios: $0.0 + 0.018 + 0.016 + 0.010 = 0.044$.

Solution 8

Question 8 (c)

Genes, diseases, and outcomes. ... [Intro text from 8(a) omitted for brevity] ...

What is the probability of paralysis ($P = 1$)?

Solution 8 (c)

We find the marginal probability $P(P = 1)$ by marginalizing over its parent, D .

$$P(P = 1) = \sum_{d \in \{0,1\}} P(D = d, P = 1)$$

Using the Law of Total Probability and the network factorization:

$$P(P = 1) = P(P = 1 \mid D = 0)P(D = 0) + P(P = 1 \mid D = 1)P(D = 1)$$

From part (b), we know $P(D = 1) = 0.044$. Therefore, $P(D = 0) = 1 - P(D = 1) = 1 - 0.044 = 0.956$. Now, we plug in all values from our CPTs and previous calculations:

$$\begin{aligned} P(P = 1) &= (0.01) \cdot (0.956) + (0.5) \cdot (0.044) \\ &= 0.00956 + 0.022 \\ &= 0.03156 \end{aligned}$$

$\therefore$ The probability of paralysis ($P = 1$) is 0.03156, or $\approx 3.16\%$.

Graduate Level Explanation

This is another marginal inference computation. The probability $P(D)$ computed in 8(b) acts as the prior belief (or "message") for the node D . We then use this belief to compute the marginal for P by applying the chain rule, $P(P) = \sum_D P(P \mid D)P(D)$. This two-step process demonstrates how belief propagates sequentially through the network.

Learn Fast Explanation

What's the total chance of paralysis? Again, we find the chance of *each way* it can happen and add them up.

1. (No disease) AND (Paralysis): $P(D = 0) \times P(P = 1 \mid D = 0) = (0.956) \times (0.01) = 0.00956$
2. (Have disease) AND (Paralysis): $P(D = 1) \times P(P = 1 \mid D = 1) = (0.044) \times (0.5) = 0.022$

Add those two scenarios: $0.00956 + 0.022 = 0.03156$.

Solution 8

Question 8 (d)

Genes, diseases, and outcomes. ... [Intro text from 8(a) omitted for brevity] ...

We observe that a person has paralysis. What is the probability that they have the disease?

Solution 8 (d)

We need to calculate the conditional probability $P(D = 1 \mid P = 1)$. We use Bayes' rule:

$$P(D = 1 \mid P = 1) = \frac{P(P = 1 \mid D = 1)P(D = 1)}{P(P = 1)}$$

We have all three components from our previous calculations:

- $P(P = 1 \mid D = 1) = 0.5$ (from the CPT for P)
- $P(D = 1) = 0.044$ (from part 8b)
- $P(P = 1) = 0.03156$ (from part 8c)

Now, plug them in:

$$\begin{aligned} P(D = 1 \mid P = 1) &= \frac{0.5 \times 0.044}{0.03156} \\ &= \frac{0.022}{0.03156} \\ &\approx 0.69708 \end{aligned}$$

$\therefore$ The probability that a person with paralysis has the disease is ≈ 0.697 (or 69.7)

Graduate Level Explanation

This is a classic example of diagnostic reasoning (abductive inference) in a Bayesian network. We are calculating a posterior probability $P(\text{Cause} \mid \text{Effect})$. The prior probability of disease $P(D = 1)$ was only 4.4

Learn Fast Explanation

This is Bayes' rule in action. We want $P(\text{Disease} \mid \text{Paralysis})$.

- ****Numerator:**** What's the chance of (Have disease) AND (Paralysis)? We found this in 8(c): 0.022.
- ****Denominator:**** What's the *total* chance of (Paralysis)? We found this in 8(c): 0.03156.

So, the probability is just $\frac{0.022}{0.03156} \approx 0.697$. In plain English: "Of all the ways a person could be paralyzed (which is 3.16

Solution 8

Solution 8 (e)

We need to calculate $P(G_1 = 1 \mid D = 1)$. We use Bayes' rule:

$$P(G_1 = 1 \mid D = 1) = \frac{P(D = 1, G_1 = 1)}{P(D = 1)}$$

We have the denominator $P(D = 1) = 0.044$ from part (8b). We must calculate the numerator, which is the joint probability $P(D = 1, G_1 = 1)$. To do this, we marginalize over G_2 :

$$\begin{aligned} P(D = 1, G_1 = 1) &= \sum_{g_2 \in \{0,1\}} P(G_1 = 1, G_2 = g_2, D = 1) \\ &= P(G_1 = 1)P(G_2 = 0)P(D = 1 \mid G_1 = 1, G_2 = 0) \\ &\quad + P(G_1 = 1)P(G_2 = 1)P(D = 1 \mid G_1 = 1, G_2 = 1) \end{aligned}$$

Using values from the CPTs:

$$\begin{aligned} P(D = 1, G_1 = 1) &= (0.1)(0.8)(0.2) + (0.1)(0.2)(0.5) \\ &= 0.016 + 0.010 \\ &= 0.026 \end{aligned}$$

Now we can solve Bayes' rule:

$$\begin{aligned} P(G_1 = 1 \mid D = 1) &= \frac{0.026}{0.044} = \frac{26}{44} = \frac{13}{22} \\ &\approx 0.5909 \end{aligned}$$

$\therefore$ The probability that a person with the disease has the mutant G_1 gene is $\frac{13}{22} \approx 0.591$ (or 59.1

Graduate Level Explanation

This is another diagnostic inference, $P(G_1 = 1 \mid D = 1)$. We cannot use the simpler form $P(D = 1 \mid G_1 = 1)P(G_1 = 1)$ because $P(D = 1 \mid G_1 = 1)$ is not a value in our CPTs, as D also depends on G_2 . We must first compute the joint $P(D = 1, G_1 = 1)$ by marginalizing out G_2 . This demonstrates an inference query that "skips" a node in the graph, as we are reasoning from D back to its parent G_1 .

Learn Fast Explanation

We want $P(\text{Mutant } G_1 \mid \text{Disease})$.

- ****Denominator:**** What's the total chance of having the disease? From 8(b), it's 0.044.
- ****Numerator:**** What's the chance of (Mutant G_1) AND (Disease)? We find this by adding the two ways it can happen: 1. (Mutant G_1) AND (Normal G_2) AND (Disease): $(0.1 \times 0.8 \times 0.2) = 0.016$
2. (Mutant G_1) AND (Mutant G_2) AND (Disease): $(0.1 \times 0.2 \times 0.5) = 0.010$ Total for numerator = $0.016 + 0.010 = 0.026$.

The answer is $\frac{0.026}{0.044} \approx 0.591$. The chance of having the mutant gene G_1 was only 10

Solution 9

Question 9 (a)

A character-level RNN for text generation. Download the file `rnn-gen.zip` from the course webpage; it contains a Jupyter notebook, `text-gen-rnn.ipynb`, as well as data and helper files. The notebook is adapted from an excellent blog post by Andrej Karpathy, *The unreasonable effectiveness of recurrent neural networks*. It fits an RNN (more precisely, an LSTM) to the text of Shakespeare's plays and then uses this to generate more text, one character at a time. Work through the notebook and try to understand what is going on. There are just three small questions to answer. How many characters are we considering, and what is the integer code for 'A'?

Solution 9 (a)

These answers are found by running the notebook cells and inspecting the `char_to_index` dictionary created from the `chars.npy` file.

1. **Number of characters:** This is the length of the `chars` array or the `char_to_index` dictionary. The question for part (c) confirms that the total number of unique characters in the vocabulary is 66.
2. **Integer code for 'A':** This depends on the exact contents and order of the `chars.npy` file, which is not provided in the script. To find this, one would run the cell that defines `char_to_index` and then execute :

```
1 # This cell must be run after the cell defining char_to_index
2 print(f"Total characters in vocabulary: {len(char_to_index)}")
3 print(f"Integer code for 'A': {char_to_index['A']}")
```

Verification

The number of characters (66) is confirmed by cross-referencing part (c) of the question. The method to find the code for 'A' (`char_to_index['A']`) is incorrect. The exact value for 'A' cannot be determined from the provided script alone.

∴ There are **66** unique characters in the vocabulary. The integer code for 'A' is found by running `print(char_to_index['A'])` after loading the `chars.npy` file, as its specific value depends on the order of characters in that file.

Graduate Level Explanation

The 66 characters form the vocabulary of this language model. This process, known as tokenization, maps discrete symbols (characters) to numerical representations (integers). These integers serve as indices for an embedding layer, which translates the sparse, 66-dimensional one-hot representation of a character into a dense, lower-dimensional vector. The size of this vocabulary dictates the dimensionality of the model's input embedding table and its final output (softmax) layer.

Learn Fast Explanation

Think of the 66 characters as the total number of "keys" on the model's typewriter (e.g., 'a', 'b', 'c', 'A', 'B', '!', ' ', '\n').

- **Number of characters:** 66 is just the total count of these unique keys.
- **Integer code for 'A':** This is like the ID number for the 'A' key. The computer can't read 'A', but it can read the ID number (like 11, or 34, or whatever it is in the file). We just need to run the code to see what ID number was assigned to 'A'.

Solution 9

Question 9 (b)

A character-level RNN for text generation. Download the file `rnn-gen.zip` from the course webpage; it contains a Jupyter notebook saved as `.py`, `text-gen-rnn.ipynb`, as well as data and helper files. The notebook is adapted from an excellent blog post by Andrej Karpathy, *The unreasonable effectiveness of recurrent neural networks*. It fits an RNN (more precisely, an LSTM) to the text of Shakespeare's plays and then uses this to generate more text, one character at a time.

Work through the notebook and try to understand what is going on. There are just three small questions to answer.

What is the length of the corpus in characters?

Solution 9 (b)

∴

Solution 9

Question 9 (c)

A character-level RNN for text generation. Download the file `rnn-gen.zip` from the course webpage; it contains a Jupyter notebook saved as `.py`, `text-gen-rnn.ipynb`, as well as data and helper files. The notebook is adapted from an excellent blog post by Andrej Karpathy, *The unreasonable effectiveness of recurrent neural networks*. It fits an RNN (more precisely, an LSTM) to the text of Shakespeare's plays and then uses this to generate more text, one character at a time.

Work through the notebook and try to understand what is going on. There are just three small questions to answer.

The input and output size for the RNN are set to 66. Why is this?

Solution 9 (c)

∴
